Wavefront set and reduction to depth-0

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https://chengchiangtsai.github.io/Hangzhou.pdf

Lusztig–Spaltenstein induction

- $F/\mathbb{Q}_p$ a p-adic field. F^{ur} maximal unramified, F^s separable closure.
- $G/_F$ connected reductive.
- Consider $X \in \mathfrak{g}(F)$ semisimple. $H := Z_G(X)$ a twisted Levi subgroup.
- For $E \in \{F, F^{ur}, F^s\}$ and any subset $S_H \subset \mathfrak{h}^{\mathrm{nil}}(E)$ closed and $\mathrm{Ad}(H(E))$-invariant. Define

$$\mathrm{Ind}_{H,X}^{G}(S_H) := \overline{(E^\times)^2 \cdot \mathrm{Ad}(G(E))(X + S_H)} \cap \mathfrak{g}^{\mathrm{nil}}(E) \subset \mathfrak{g}^{\mathrm{nil}}(E).$$

- **Lemma 1.** When $E = F^s$, for orbit $\mathcal{O}_H \subset \mathfrak{h}^{\mathrm{nil}}(F^s)$ we have

$$\mathrm{Ind}_{H,X}^{G}(\overline{\mathcal{O}_H}) = \overline{\mathcal{O}_G}$$

  where $\mathcal{O}_G$ is the Lusztig–Spaltenstein induction, namely the unique orbit in $\mathfrak{g}^{\mathrm{nil}}(F^s)$ such that $\mathcal{O}_G \cap (\mathcal{O}_H + \mathfrak{u}_P) \subset \mathcal{O}_H + \mathfrak{u}_P$ is open dense for every parabolic P with Levi H.
- Unsurprisingly, the same is true for GL_n over any local field E.

$\mathrm{Gal}(F^s/F)$ is too big

$\mathrm{Ind}_{H,X}^{G}(S_H) := \overline{(E^\times)^2 \cdot \mathrm{Ad}(G(E))(X + S_H)} \cap \mathfrak{g}^{\mathrm{nil}}(E)$.

- **Lemma 2.** When $E = F^{ur}$ and $S_H = \overline{\mathcal{O}_H}$ is the (analytic) closure of a single orbit, $\mathrm{Ind}_{H,X}^{G}(\overline{\mathcal{O}_H})$ contains an orbit in the Lusztig–Spaltenstein induction.
- Lemma 2 doesn't hold for $E = \mathbb{R}$. Instead,

  Lemma 3. When $E = \mathbb{R}$, $\mathrm{Ind}_{H,X}^{G}(\overline{\mathcal{O}_H}) = \bigcup \overline{\mathcal{O}_G^i}$, for some $\mathcal{O}_G^i$ in a single geometric orbit, possibly smaller than the LS induction.
- Neither Lemma 2 nor 3 holds for $E = F$; the maximal orbits in $\mathrm{Ind}_{H,X}^{G}(\overline{\mathcal{O}_H})$ may live in **incomparable** geometric orbits.
- **Example.** (Tsai '23, ChatGPT–Tsai '26) Let $p \geq 13$ and consider
  - an unramified $H := U_2 \times U_1^4 \hookrightarrow \mathrm{Sp}_{12} =: G$,
  - $X \in \mathfrak{g}(F)$, with $Z_G(X) = H$ and non-zero root values in $\mathcal{O}_{F_2}^\times$,
  - $\mathcal{O}_H \subset \mathfrak{h}(F)$ a regular nilpotent orbit specifying a vertex

  $x \in \mathcal{B}(H) \subset \mathcal{B}(G)$, such that $\mathbb{G}_x \cong \mathrm{Sp}_6 \times \mathrm{Sp}_6$.
  Then $\mathrm{Ind}_{H,X}^{G}(\overline{\mathcal{O}_H})$ is the union of closures of 4 orbits of Jordan type (66) and 8 orbits of Jordan type (822).

Asymptotic cone and wavefront

$\mathrm{Ind}_{H,X}^{G}(\overline{\mathcal{O}_H})$ is the union of closures of orbits of Jordan type (66) and (822).

- **Theorem.** (DeBacker–Tsai) The following are equivalent:
  (a) $\mathcal{O}_G$ is a maximal orbit in $\mathrm{Ind}_{H,X}^{G}\overline{\mathcal{O}_H}$.
  (b) $\mathcal{O}_G$ is a maximal orbit such that $\mathrm{Ad}(G)(X+\mathcal{O}_H)$ intersects a Slodowy slice for $\mathcal{O}_G$.
  (c) $\mathcal{O}_G$ is a maximal orbit in the Shalika germ expansion of $\mu_{X+\mathcal{O}_H}^{G}$.
  (d) $\mathcal{O}_G$ is a maximal orbit in the local expansion of $\hat{\mu}_{X+\mathcal{O}_H}^{G}$.
- **Corollary.** There are examples of rational wavefront sets of
  - Positive-depth supercuspidal representations of Sp_{2n}, $n \geq 6$
  - Depth-0 tempered (non-discrete) representations of Sp_{2n}, $n \geq 6$.

  to consist of orbits of Jordan type $(2n-4,2,2)$ and $(2n-6,6)$. These happen for arbitrarily large p.
- Similarly for split SO_{2n+1} ($n \geq 6$), split SO_{2n} ($n \geq 8$), and U_n ($n \geq 6$).

Wavefront set and reduction to depth-0

- In Adler–Fintzen–Mishra–Ohara '24, as well as in Dat–Fintzen '26, we have twisted Levi subgroup $H \hookrightarrow G$ and certain bijection

$$\begin{array}{ccc} \pi_H & \mapsto & \pi_G \\ \cap & & \cap \\ \mathrm{Irr}(H)_{\mathfrak{s}_H^0} & \xrightarrow{\sim} & \mathrm{Irr}(G)_{\mathfrak{s}_G} \end{array}$$

  where $\mathfrak{s}_H^0$ is a depth-0 (Bernstein) block and $\mathfrak{s}_G$ is some corresponding positive-depth block.
- These bijections **depend** on choices.
- **Conjecture.** Subject to conditions on the choices, F and G, we have

$$\mathrm{Ind}_{H,X}^G(\mathrm{WF}(\pi_H)) = \mathrm{WF}(\pi_G).$$

  For some X determined by the block.

Dat–Fintzen reduction, Galois side

Conj. $\mathrm{Ind}_{H,X}^{G}(\mathrm{WF}(\pi_H)) = \mathrm{WF}(\pi_G)$.

- From now on, suppose G is tamely ramified, quasi-split, and p does not divide the order of the absolute Weyl group of G.
- $P_F \lhd W_F$ wild inertia in the Weil group.
- In this hour, ${}^L G := G^\vee \rtimes (W_F/P_F)$.
- Let $\varphi : W_F \times \mathrm{SL}_2(\mathbb{C}) \to {}^L G$ be a Langlands parameter.
- **Lemma.** $H^\vee := Z_{G^\vee}(\varphi(P_F))$ is a Levi subgroup, and $\varphi(P_F) \subset A^\vee := Z(H^\vee)^\circ$.
- We have W_F/P_F acting on $H^\vee$ via φ. Fix a pinning for $H^\vee$ and consider the corresponding pinned (outer) action of W_F/P_F on $H^\vee$. We may form ${}^L H := H^\vee \rtimes (W_F/P_F)$.
- Dually we get a quasi-split $H/_F$. Also $A \cong H/H^{\mathrm{der}}$.
- **Proposition.** (Dat–Fintzen) There exists (non-unique) embeddings

$$ {}^L\iota : {}^L H \hookrightarrow {}^L G $$

extending $H^\vee \hookrightarrow G^\vee$ such that φ factors through ${}^L\iota$.

More on Galois side

$H^\vee = Z_{G^\vee}(\varphi(P_F))$, $A^\vee := Z(H^\vee)^\circ$, $\quad \varphi : W_F \times \mathrm{SL}_2(\mathbb{C}) \xrightarrow{\varphi_H} {}^L H \xhookrightarrow{{}^L\iota} {}^L G$

- Let $\phi = \varphi|_{P_F} : P_F \to A^\vee$. Extend it (non-uniquely) to a Langlands parameter $\tilde{\phi} : W_F \to {}^L A$. Then the "central twist" $\varphi'_H := \varphi_H \cdot \tilde{\phi}^{-1}$ is trivial on P_F.
- We have
$$Z_{G^\vee}(\varphi) = Z_{H^\vee}(\varphi_H) = Z_{H^\vee}(\varphi'_H).$$
- For any $\rho \in \mathrm{Irr}(\pi_0(Z_{G^\vee}(\varphi)))$, we have its central character
$$\alpha_\rho \in \mathrm{Hom}(\pi_0(Z(H^\vee)^{W_F}), \mathbb{C}^\times) \cong H^1(F, H)$$
This gives an inner twist $H_\rho \hookrightarrow G_\rho$ of our chosen $\iota : H \hookrightarrow G$.
- LLC predicts matching representations
$$(\pi^H_{\varphi,\rho}, \pi^G_{\varphi,\rho}) \in \mathrm{Irr}(H_\rho) \times \mathrm{Irr}(G_\rho).$$
Same for rigid inner twists.
- Our goal is, roughly,
$$\mathrm{WF}(\pi^G_{\varphi,\rho}) \stackrel{?}{=} \mathrm{Ind}^{G_\rho}_{H_\rho, X} \mathrm{WF}(\pi^H_{\varphi,\rho}).$$

X^* and the Whittaker data

$\mathrm{WF}(\pi^G_{\varphi,\rho}) \overset{?}{=} \mathrm{Ind}^{G_\rho}_{H_\rho,X} \mathrm{WF}(\pi^H_{\varphi,\rho})$.

- The extension $\tilde{\phi} : W_F \to {}^L A$ is a character $\chi_{\tilde{\phi}} : A(F) \to \mathbb{C}^\times$.
- **Lemma.** The restriction $\chi_\phi := \chi_{\tilde{\phi}}|_{A(F)_{0+}}$ to the subset $A(F)_{0+}$ of pro-p elements depend only on ϕ.
- Fix $\psi : F \to \mathbb{C}^\times$.
  Lemma. There exists $X^* \in \mathfrak{a}^*(F) \cong \mathfrak{g}^*(F)^H$ such that $Z_G(X^*) = H$ and
  $$\chi_\phi(a) = \psi(\log(a), X^*), \quad \forall a \in A(F)_{0+}.$$
  Here we assume log "converges well".
- Suppose regular nilpotent elements $e_H \in \mathfrak{h}^*(F)$ and $e_G \in \mathfrak{g}^*(F)$ are given.
  Theorem. (DeBacker–Tsai) There is a unique $G(F)$-conjugacy class of $\iota : H \hookrightarrow G$ (dual to $H^\vee \hookrightarrow G^\vee$ over F^s) such that
  $$e_G \in \mathrm{Ind}^G_{H,X^*}(\overline{\mathrm{Ad}(H(F))e_H})$$
  When $p \gg \mathrm{rank}_{F^s}(G)$, this conjugacy class of ι is independent of X^*.

Conjectures

$\mathrm{Ind}_{H,X}^{G}(S_H) := \overline{(E^\times)^2 \cdot \mathrm{Ad}(G(E))(X + S_H)} \cap \mathfrak{g}^{\mathrm{nil}}(E)$.

- We are given $\varphi : W_F \times \mathrm{SL}_2(\mathbb{C}) \to {}^L G$ and $\rho \in \mathrm{Irr}(\pi_0(Z_{G^\vee}(\varphi)))$.
- **Conjecture A.** Let

$$(\pi_{\varphi,\rho}^H, \pi_{\varphi,\rho}^G) \in \mathrm{Irr}(H_\rho) \times \mathrm{Irr}(G_\rho)$$

be as predicted by LLC. When p is larger than some constant in terms of $\mathrm{rank}_{F^s}(G)$

$$\mathrm{WF}(\pi_{\varphi,\rho}^G) = \mathrm{Ind}_{H_\rho, X^*}^{G_\rho}(\mathrm{WF}(\pi_{\varphi,\rho}^H)) \qquad (**)$$

for any choice of ${}^L\iota$, $\tilde{\phi}$ and X^* above.
- **Conjecture B.** Same with $(\pi_{\varphi,\rho}^H, \pi_{\varphi,\rho}^G)$ given by any choice of Dat–Fintzen equivalences.
- **Conjecture C.** Same with $(\pi_{\varphi,\rho}^H, \pi_{\varphi,\rho}^G)$ given by a specific kind of Adler–Fintzen–Mishra–Ohara equivalence.
- Same with Ma–Wang–Yu equivalence?
- **Remark.** It is unclear if (**) holds just when $p \nmid |W_G|$; we expect $\subset$ to always hold.

Partial results

Conj. $\mathrm{WF}(\pi^G_{\varphi,\rho}) = \mathrm{Ind}^{G_\rho}_{H_\rho, X^*}(\mathrm{WF}(\pi^H_{\varphi,\rho}))$.

- We assume p is very large.
- **Theorem.** (DeBacker–Tsai) The validity of the conjectures depends only on ϕ but otherwise not on X^*.
- **Theorem.** (Ohara–Tsai) For a specific non-empty sub-families of AFMO equivalences, (**) is true for regular nilpotent orbits.
- **Remark.** ChatGPT 6 Astra Ultra effort was run for 3-4 hours. The conjecture was neither proved nor disproved, but ChatGPT seems to prove that $\subset$ is true (when p is very large).
- **Remark.** For mod-ℓ representations it is possible to work with Brauer characters, their local character expansions, and a corresponding notion of wavefront set. We expect everything so far to work for that as well.

The end. Thank you!